

Oxide AI — project handoff (updated)

Snapshot: 17 September 2026 UTC / 17 September 2026 Pacific morning. Owner: Spart / sparticle62ops. Supersedes PROJECT-HANDOFF-2026-09-17.md (written before the items in Section 1 below).

Private recovery document. Do not publish the complete recovery archive, private transcripts, or corpus files to GitHub.

1. What changed since the last handoff

- **Pushed to GitHub.** Draft PR #2: <https://github.com/Sparticle62ops/oxide-ai/pull/2> — branch research/stacked-depth-wip-20260917, commit 5cdd9fb, 17 files (stacked-depth source, tests, study runner, notes).

- **Attribution note:** the connected GitHub integration cannot set custom author/committer identity — commits it makes are recorded under the connected Tasklet-linked account, not literally sparticle62ops on both fields. On 2026-09-17 05:28 UTC you explicitly authorized this, on condition that sparticle62ops is retained as credited co-author in the commit message/trailer. That is what was done for this push. The fully sparticle62ops-identity local commits (edea5454. . .) remain separately preserved in the recovery bundle if you later push them yourself with your own Git client.

- **Stage B (real multilayer stacking) is now implemented and passing**, superseding the incomplete/failing snapshot recorded in depth-project/STACKED-WIP.md. Full detail in depth-evidence/stage-b/RESULT.md:

- Build exit 0, full regression suite exit 0: **47 tests passed, 0 failed**, across 14 test binaries (10 prior integration suites + 4 new stacked-model tests + 1 new allocation test). - PSSALayerV2: :new_with_depth supports depth 1-32; depth-1 still saves the original V7 checkpoint format; depth>1 saves the new V8 format. V5/V6/V7 loading compatibility is retained. - Covers depth-1 init parity, depth-4 train/inference equivalence, non-vacuous boundary adjoints and finite differences (including populated-memory cases), one shared optimizer clock, V8 round-trip and corrupted-input rejection, depth-2 learning smoke test, and warmed no-allocation hot paths. - Not yet done: an exhaustive per-parameter finite-difference matrix at every depth-2/4 block (existing depth-1 coverage remains the strict baseline), and no training run has happened yet.

- **The width/depth comparison runner is implemented but unexecuted.**

depth-project/examples/width_depth_study.rs plus contract in depth-evidence/STUDY-RUNNER-CONTRACT.md / depth-evidence/study-runner/RESULT.md. It supports resumable, fingerprinted, checkpointed runs against the frozen protocol manifest schema (width, depth, epochs=4, seed=42, chunk=64, etc.), with preflight-only validation, periodic/epoch checkpoints, diagnostics, and final generations. Its own unit test has not yet been compiled/run — that's the next concrete step, followed by the actual four training runs (width64/depth1, width256/depth1, width64/depth2, width64/depth4).

- Added depth-project/BUILDING.md (reproducible Rust setup/build/cache guidance) and depth-project/rust-toolchain.toml pinning **Rust 1.90.0**. Per your request, the pinned toolchain version is tracked in the repo so setup is reproducible — but the actual compiler installation is not committed (it's OS/CPU-specific and would bloat the repo without being portable); caching ~/.rustup, ~/.cargo, and target/ between sessions is the effective way to skip repeated install/compile time.

2. Still true from before (unchanged)

- **Broadly coherent, logical language generation has NOT been achieved.** This remains the success bar — not compilation, not lower perplexity, not layer count.
- Verified training evidence (width64/depth1, unchanged from prior handoff):
- Init: NLL 7.9479712, perplexity 2829.83, accuracy 0.0264%. - Full pass 1: perplexity 175.9979, accuracy 11.554%. - Longer control epoch1/2/3: perplexity 175.233 / 134.529 / 114.801; accuracy 11.602% / 14.074% / 15.686%. **Only 3 of 4 planned passes completed** for that run — no epoch4/final checkpoint exists for it. - Test split still not evaluated. Sample degeneration persists (e.g., "the purpose of a scientific experiment is" → "the sodium and the state... United States...").
- User's binding order: **fix/train the model until language is reproducibly coherent → then CLI redesign → then full WGPU auto-detect/interactive backend support.** No CLI or GPU work has started, correctly.
- The frozen comparison protocol (experiments/WIDTH-DEPTH-PROTOCOL.md) is unchanged: 4 configurations, 4 full passes each, same seed/tokenizer/corpus/schedule, final-epoch primary comparison, boundary-gradient diagnostics, full generation review, one-seed result treated as pilot only.

3. File map (persistent root

/tasklet/threads/a_rkbhvdvaz9a63sqhx7mr/work)

- project/ — known-good depth-1 baseline, untouched.
 - depth-project/ — active source; now includes accepted Stage B stacking code.
 - depth-evidence/stage-a-complete/, depth-evidence/stage-b/ — audit trails, build/test logs, exit codes for each accepted stage.
 - depth-evidence/study-runner/, depth-evidence/STUDY-RUNNER-CONTRACT.md — comparison-runner implementation record.
 - experiments/ — training records, checkpoints, fixed generations, the frozen protocol.
 - publication/ — attributed bundles, mailbox patches, GitHub preflight/status checks.
 - oxide-ai-handoff/ — original imported project and private prior context.
- /tmp remains ephemeral; a reset already destroyed one build cache this session. Only the work directory above is guaranteed durable.

4. Resume order

1. Compile and run the study-runner's own unit test (`cargo test --example width_depth_study`), not yet done.
2. Run `preflight_only` for all four manifests; confirm 840,951 transitions/pass and matching fingerprints before spending compute.
3. Execute the four training configurations sequentially: width64/depth1 (fresh baseline — the old 3-pass run doesn't count), width256/depth1, width64/depth2, width64/depth4.
4. Compare final-epoch perplexity/accuracy, boundary-gradient diagnostics, throughput, and full unselected generations across all four. Judge coherence directly, not just perplexity.
5. Only after a config shows real, reviewer-checked coherent/logical output improvement: begin CLI redesign, then WGPU auto-detect/interactive backend support.
6. Keep pushing tested source to GitHub (via the current connection, Tasklet-linked identity + sparticle62ops co-author credit, unless you push yourself for pure identity) after each accepted milestone. Never push the private recovery archive, transcripts, or corpus wholesale.

5. Open items needing your input eventually

- Whether to eventually merge draft PR #2 into main, or keep iterating on the branch until the width/depth study finishes.
- Whether you want to push the pure-particle62ops-identity local commits yourself at some point (recovery bundle has them ready), in addition to the Tasklet-attributed PR.